A SCATTERED LINEAR ORDER OF SCOTT SENTENCE COMPLEXITY PI^IN_LAMBDA FOR EVERY LIMIT ORDINAL LAMBDA

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Editorial note. The displayed definition at source line 1403 prints the summation variable as delta_n; the article's own later lines 1410 and 1419 use the same fundamental sequence consistently, and the parallel construction at 1435 repeats the line. The source text is retained verbatim and no symbol was corrected or restructured.

Editorial note. The retained excerpt at source lines 45-53 ends in the middle of the article's sentence 'While the latter is not an ...'; the sentence continues at line 54, which points at the invariant table and is not extracted.

Editorial note. The article's Z-power back-and-forth lemma (source lines 1147-1266) is delivered with its complete proof because the optimality argument depends on it; it is already prepared in the corpus as problem 141394 and is not a newly counted target here.

Editorial note. The two citations inside the extracted proof (Alvir-Rossegger on scattered orders; Montalban on classes of structures with no intermediate isomorphism problems) are resolved by the retained bibliography fragment listed with this task.

Editorial note. The retained environments print with this package's own numbering rather than the article's section-based numbering; original labels are kept in the span metadata.

Editorial scope. One original exact result: the existence of a scattered linear order of Scott sentence complexity Pi^in_lambda for every limit ordinal, with the article's own complete Z-power back-and-forth lemma as source-local support.

Allowed standard prerequisites (editorial).

- Fundamental sequences and countable limit ordinals. Every countable limit ordinal lambda has a strictly increasing cofinal sequence (delta_n) of ordinals below lambda.
- Scattered orders, Hausdorff rank and Z^alpha. Standard Hausdorff analysis of scattered linear orders and the cited bound that a scattered order of Hausdorff rank below lambda has Scott sentence complexity below Pi^in_lambda (Alvir-Rossegger, cited at source line 1416); the back-and-forth behaviour of Z^alpha-multiples is the article's own lemma, delivered here with its complete proof.
- Parameter-definable interval lower bound. The cited criterion that a Delta^in_0-definable interval over parameters cannot exceed the Scott rank of the ambient structure (Montalban, cited at source line 1421); used only for the optimality step.

In this article we give a comprehensive Scott analysis of the class of linear orderings using three definitions of rank. Given a structure $\mathcal{A}$ the *(parameterless) Scott rank* $SR(\mathcal{A})$, is the least α, such that the automorphism orbits of all tuples in $\mathcal{A}$ are $\Sigma^{\text{in}}_{\alpha}$

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definable; the *parameterized Scott rank* $pSR(\mathcal{A})$, is the least ordinal α such that there is a parameter $\bar{p} \in A^{<\omega}$ such that $SR((\mathcal{A}, \bar{p})) = \alpha$; and the *Scott sentence complexity* of $\mathcal{A}$, $SSC(\mathcal{A})$ is the least of the complexities Π^{in}_α, $\Sigma^{\text{in}}_\alpha$, $d\text{-}\Sigma^{\text{in}}_\alpha$ such that $\mathcal{A}$ has a Scott sentence of one of these complexities. While the latter is not an

Given these base case examples of linear orders of small Scott sentence complexity, we now devise a method of systematically using these examples to fill in most of the larger Scott sentence complexities. The examples are quite simple; we consider orders of the form $\mathbb{Z}^\alpha \cdot \mathcal{L}$ where $\mathcal{L}$ is one of our previously constructed examples. The more difficult work is determining exactly of what Scott sentence complexity these orders are.

This will be proven by demonstrating an upper and lower bound on the complexity of the Scott sentence. We begin with proving the lower bound. In the proof of the below lemma we use ζ_α to denote the unique up to isomorphism initial segment of $\mathbb{Z}^\alpha$. It is easy to check that ζ^*_α is the unique end segment of $\mathbb{Z}^\alpha$ and that $\zeta^*_{\alpha+1} = \zeta_\alpha + \mathbb{Z}^\alpha \cdot \omega$.

This covers every case except for some possibilities that lie close to limit ordinals. The following results fill most of those gaps.

Proposition 0.1. *For any limit ordinal λ, there is a scattered linear order of Scott sentence complexity Π^{in}_λ.*

Lemma 0.2. *For the sake of organization, consider the following ordinal indexed propositions.*

- (A_α) *For any $\mathcal{K}$ and $\mathcal{L}$, $\mathbb{Z}^\alpha \cdot \mathcal{K} \leq_{2\alpha} \mathbb{Z}^\alpha \cdot \mathcal{L}$.*
- (B_α) *For any $\mathcal{K}$ and $\mathcal{L}$ with $|\mathcal{K}| \geq |L|$, $\mathbb{Z}^\alpha \cdot \mathcal{K} \leq_{2\alpha+1} \mathbb{Z}^\alpha \cdot \mathcal{L}$.*
- (C_α) *For any $\mathcal{K}$ and any $\mathcal{L}$ without a last element, $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}) \leq_{2\alpha+2} \mathbb{Z}^\alpha \cdot (\omega + \mathcal{L})$.*

For any countable ordinal α, A_α, B_α and C_α are true.

Proof. The proof is by induction with the statements A_0 and B_0 trivially true. We start by showing that B_α implies $A_{\alpha+1}$.

Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L} = \mathbb{Z}^\alpha \cdot \mathbb{Z} \cdot \mathcal{L}$. We will find a tuple $b_0 \dots b_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K} = \mathbb{Z}^\alpha \cdot \mathbb{Z} \cdot \mathcal{K}$ such that $(\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}, \bar{a}) \leq_{2\alpha+1} (\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}, \bar{b})$. Pick a point b_0 arbitrarily. Given b_j, we pick b_{j+1} as follows. If a_j and a_{j+1} are in the same copy of $\mathbb{Z}^{\alpha+1}$, we can pick b_{j+1} such that $(a_j, a_{j+1}) \cong (b_j, b_{j+1})$. If not, we pick b_{j+1} arbitrarily in the copy of $\mathbb{Z}^\alpha$ that is the successor of the successor of the copy of b_j. In other words, we pick b_{j+1} with $(b_j, b_{j+1}) \cong \zeta^*_\alpha + \mathbb{Z}^\alpha + \zeta_\alpha$.

To show that our choice of $\bar{b}$ is correct, we have to confirm the following relations between non-isomorphic intervals.

1. $(b_j, b_{j+1}) \cong \zeta^*_\alpha + \mathbb{Z}^\alpha + \zeta_\alpha \geq_{2\alpha+1} \zeta^*_\alpha + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$, where $\mathcal{L}_1$ is an infinite order,
2. $(-\infty, b_0) \cong \mathbb{Z}^\alpha \cdot \mathcal{K}_1 + \zeta_\alpha \geq_{2\alpha+1} \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$, where $\mathcal{L}_1$ and $\mathcal{K}_1$ are some initial segments of the $\mathbb{Z}^\alpha$, and therefore infinite, or
3. $(b_n, \infty) \cong \zeta^*_\alpha + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 \geq_{2\alpha+1} \zeta^*_\alpha + \mathbb{Z}^\alpha \cdot \mathcal{K}_1$, where $\mathcal{L}_1$ and $\mathcal{K}_1$ are some final segments of the $\mathbb{Z}^\alpha$, and therefore infinite.

It follows immediately from B_α that all of these back-and-forth relations hold.

We now show that B_α implies C_α. So assume that $\mathcal{K}$ and $\mathcal{L}$ have no last element and that B_α holds. Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{L})$. Without loss of generality, a_0 is in the first copy of $\mathbb{Z}^\alpha$. We will find a tuple $b_0 \dots b_n$ in $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K})$ such that $(\mathbb{Z}^\alpha \cdot (\omega + \mathcal{L}), \bar{a}) \leq_{2\alpha+1} (\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}), \bar{b})$. For the a_j in the initial $\mathbb{Z}^\alpha \cdot \omega$, the b_j are picked isomorphically. We now define the rest of the b_j by induction so that all of them are in the initial $\mathbb{Z}^\alpha \cdot \omega$. If a_j is in the same $\mathbb{Z}^\alpha$ block as a_{j+1} or a_{j+1} is in the succesor of a_j's block, define b_{j+1} so that $(b_j, b_{j+1}) \cong (a_j, a_{j+1})$. Otherwise, define b_{j+1} as being some element in the successor of the successor of the $\mathbb{Z}^\alpha$ block of b_j. Not including isomorphic intervals, we only need to check the following cases.

(1) $(b_j, b_{j+1}) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha + \zeta_\alpha \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$ where $\mathcal{L}_1$ is infinite,
(2) $(b_n, \infty) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}) \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1$, where $\mathcal{L}_1$ is some final segment of $\omega + \mathcal{L}$, and thus infinite.

All of these relations follow from B_α, as desired.

To complete the proof of the successor step of the induction, we show that C_α and $A_{\alpha+1}$ imply $B_{\alpha+1}$.

Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}$. These points belong to at most n distinct copies of $\mathbb{Z}^{\alpha+1}$. Because $|\mathcal{K}| \geq |\mathcal{L}|$, we can match each chosen copy of $\mathbb{Z}^{\alpha+1}$ to one in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}$. Furthermore, if there is a copy of $\mathbb{Z}^{\alpha+1}$ between two of the chosen copies in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}$, we can assure that the corresponding copies in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}$ are also not successors. Also, if a_n is not in the last copy of $\mathbb{Z}^{\alpha+1}$, we can assure that neither is b_n and, similarly for a_0 and b_0 in the first copy. In particular, not including the isomorphic intervals, we only need to check the following cases.

(1) $(b_j, b_{j+1}) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 + \zeta_{\alpha+1} \geq_{2\alpha+2} \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(2) $(b_n, \infty) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 \geq_{2\alpha+2} \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(3) $(-\infty, b_0) \cong \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 + \zeta_{\alpha+1} \geq_{2\alpha+2} \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(4) $(-\infty, b_0) \cong \zeta_{\alpha+1} \geq_{2\alpha+2} \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some order $\mathcal{L}_1$,
(5) $(b_n, \infty) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 \geq_{2\alpha+2} \zeta_{\alpha+1}^*$ for some order $\mathcal{K}_1$.

The first three cases are handled immediately by $A_{\alpha+1}$. By symmetry it is sufficient to show case (5). However, this is the same as showing that

$$\zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \omega \leq_{2\alpha+2} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot (\omega + \mathbb{Z} \cdot \mathcal{K}_1).$$

This follows directly from C_α.

At last we consider the remaining limit levels. We already have seen that B_λ implies C_λ. The statement A_λ follows immediately from A_γ for $\gamma < \lambda$. A bit more nuanced is the issue of B_λ. It follows from A_λ and C_γ for $\gamma < \lambda$. Analogous to the proof in the successor case, we analyze the possibilities:

(1) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{L}_1 + \zeta_\lambda$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(2) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 \geq_\lambda \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{L}_1$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(3) $\mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \mathbb{Z}^\lambda \cdot \mathcal{L}_1 + \zeta_\lambda$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(4) $\mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \zeta_\lambda$ for some order $\mathcal{K}_1$,
(5) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 \geq_\lambda \zeta_\lambda^*$ for some order $\mathcal{K}_1$.

The first three cases are handled immediately by A_λ. By symmetry it is clear that it is enough to show case 5. However, this is the same as showing that for all $\gamma < \lambda$,

$$\zeta_\gamma^* + \mathbb{Z}^\gamma \cdot (\omega + \mathbb{Z} \cdot \zeta_\lambda^*) \leq_\gamma \zeta_\gamma^* + \mathbb{Z}^\gamma \cdot (\mathbb{Z} \cdot \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1),$$

which follows from the C_γ for $\gamma < \lambda$.

Therefore, for any countable ordinal α, A_α, B_α and C_α all hold. □

Proof. Let $(\delta_n)_{n\in\omega}$ be a fundamental sequence for λ. We show that

$$\mathcal{L}_\lambda := \sum_{i\in\omega} i + \mathbb{Z}^{\delta_n}$$

is of the desired complexity.

First note that it indeed does have a Π^{in}_λ Scott sentence. The sentence states the following:

(1) There is exactly one 1-block isomorphic to each natural number.
(2) These 1-blocks are ordered like the natural numbers.
(3) The order between n and $n+1$ is isomorphic to $\mathbb{Z}^{\delta^n}$.

This is Π^{in}_λ because each of the items is $\Pi^{\text{in}}_{<\lambda}$. The first two statements can be expressed using finitely many alternations of quantifiers. The final statement can be expressed by the Scott sentence of $\mathbb{Z}^{\delta_n}$ relativized to the specific interval, this sentence has Scott sentence complexity below λ by [2], as the Hausdorff rank of the interval is below λ.

By Lemma 0.2, the $\mathbb{Z}^{\delta_n}$ have unbounded Scott rank below λ. As these are all Δ^{in}_0-definable over parameters in $\mathcal{L}_\lambda$, by [6, Lemma 4.3], we have that for each n, $SR(\mathcal{L}_\lambda) \geq \delta_n$, so Π^{in}_λ is indeed the optimal Scott sentence complexity. □

References

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